

Question 1: What is hypothesis testing in statistics?

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Hypothesis testing is a **statistical method** used to make decisions or inferences about a population based on **sample data**.

It helps determine whether there is enough evidence in a sample to **support or reject a claim (hypothesis)** about a population parameter.

Key Concepts:

1. **Null Hypothesis (H_0):**
The assumption that there is **no effect or no difference**.
Example: H_0 : The average sales = ₹500.
2. **Alternative Hypothesis (H_1 or H_a):**
The statement that **contradicts** the null hypothesis.
Example: H_1 : The average sales $\neq$ ₹500.

Question 2: What is the null hypothesis, and how does it differ from the alternative hypothesis?

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1. Null Hypothesis (H_0):

The **null hypothesis** is a **default assumption** that there is **no effect, no difference, or no relationship** in the population.

It represents the statement being **tested** and is assumed to be **true** until evidence suggests otherwise.

Example:

A company claims that the average daily sales = ₹500.

→ $H_0: \mu = 500$

It means we assume the average sales are ₹500 **unless data proves otherwise**.

2. Alternative Hypothesis (H_1 or H_a):

The **alternative hypothesis** is the **opposite** of the null hypothesis.

It states that there **is** an effect, difference, or relationship.

It represents what we want to **prove or find evidence for**.

Example:

Continuing the same case, we want to test if the sales are **different** from ₹500.

→ $H_1: \mu \neq 500$

Question 3: Explain the significance level in hypothesis testing and its role in deciding the outcome of a test.

1. What is Significance Level (α)?

The **significance level**, denoted by α (alpha), is the **threshold probability** used in hypothesis testing to decide whether to **reject the null hypothesis (H_0)**.

It represents the **maximum risk** you are willing to take of making a **Type I error** — that is, rejecting a true null hypothesis.

Common values:

- $\alpha = 0.05 \rightarrow 5\%$ risk
- $\alpha = 0.01 \rightarrow 1\%$ risk
- $\alpha = 0.10 \rightarrow 10\%$ risk

2. Role in Decision Making

After performing a hypothesis test, you get a **p-value**.

You compare it with the **significance level (α)** to make a decision:

Condition	Decision	Interpretation
$p\text{-value} \leq \alpha$	Reject H_0	Evidence suggests the result is statistically significant
$p\text{-value} > \alpha$	Fail to reject H_0	Not enough evidence to conclude a significant effect

Question 4: What are Type I and Type II errors? Give examples of each.

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1. Definition

When performing **hypothesis testing**, there are two possible kinds of mistakes that can occur when making a decision about the **null hypothesis (H_0)**:

Decision	Reality	Result
Reject H_0 when H_0 is true	✗	Type I Error
Fail to reject H_0 when H_0 is false	✗	Type II Error

2. Type I Error (False Positive)

- **Definition:**
A **Type I Error** occurs when we **reject the null hypothesis (H_0)** even though it is actually **true**. In other words, we conclude that an effect exists when it really does **not**.
- **Symbol:** α (alpha)
→ Same as the **significance level** (probability of making a Type I error).

Example:

A medical test says a patient **has a disease**, but in reality, they are **healthy**.
(We rejected the true H_0 : "The person is healthy.")

3. Type II Error (False Negative)

- **Definition:**

A **Type II Error** occurs when we **fail to reject the null hypothesis (H_0)** even though it is actually **false**.

In other words, we miss detecting a real effect.

- **Symbol:** β (beta)

Example:

A medical test says a patient **is healthy**, but they actually **have the disease**.
(We failed to reject the false H_0 : "The person is healthy.")

Question 5: What is the difference between a Z-test and a T-test? Explain when to use each.

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1. Overview

Both the **Z-test** and **T-test** are **statistical tests** used to compare sample data with population data — they help determine whether the **sample mean** differs significantly from the **population mean** (or from another sample).

However, they differ mainly in **sample size** and **knowledge of population standard deviation (σ)**.

2. Key Differences Between Z-test and T-test

Basis	Z-test	T-test
Population Standard Deviation (σ)	Known	Unknown (use sample standard deviation s)
Sample Size (n)	Large ($n \geq 30$)	Small ($n < 30$)
Distribution Used	Normal distribution	t-distribution
Shape of Distribution	Fixed (normal)	Changes with sample size (wider tails for small n)
When to Use	When data is normally distributed and σ is known	When σ is unknown or sample is small
Example Tests	Z-test for mean, Z-test for proportion	One-sample t-test, Two-sample t-test, Paired t-test

Question 6: Write a Python program to generate a binomial distribution with $n=10$ and $p=0.5$, then plot its histogram.

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Concept:

A **Binomial distribution** represents the number of successes in a fixed number of independent trials, where each trial has two possible outcomes (Success/Failure).

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Here:

- $n=10$ → number of trials
- $p=0.5$ → probability of success
- We'll simulate data and plot its histogram.

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
n = 10      # number of trials
p = 0.5     # probability of success
size = 1000 # number of experiments

# Generate binomial distribution data
data = np.random.binomial(n, p, size=size)

# Display mean and standard deviation
print(f"Mean: {np.mean(data):.2f}")
print(f"Standard Deviation: {np.std(data):.2f}")

# Plot histogram
plt.hist(data, bins=range(n+2), color='skyblue', edgecolor='black')
plt.title('Binomial Distribution (n=10, p=0.5)')
plt.xlabel('Number of Successes')
plt.ylabel('Frequency')
plt.show()
```

Explanation:

- `np.random.binomial(n, p, size)` → generates random numbers following a binomial distribution.
- `plt.hist()` → plots the histogram to visualize the frequency of successes.
- Mean $\approx n \times p = 10 \times 0.5 = 5$
- Standard deviation $\approx \sqrt{n \times p \times (1-p)} = \sqrt{10 \times 0.5 \times 0.5} = \sqrt{2.5} \approx 1.58$

Output (Typical Result):

- Mean: around **5.00**
- Standard Deviation: around **1.58**

- Histogram: a bell-shaped distribution centered near 5 successes.

Question 7: Implement hypothesis testing using Z-statistics for a sample dataset in Python. Show the Python code and interpret the results.

```
sample_data = [49.1, 50.2, 51.0, 48.7, 50.5, 49.8, 50.3, 50.7, 50.2, 49.6, 50.1, 49.9, 50.8, 50.4, 48.9, 50.6, 50.0, 49.7, 50.2, 49.5, 50.1, 50.3, 50.4, 50.5, 50.0, 50.7, 49.3, 49.8, 50.2, 50.9, 50.3, 50.4, 50.0, 49.7, 50.5, 49.9]
```

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1. Problem Setup

We have a **sample dataset** of product weights (in grams).

We want to test whether the **mean weight = 50 grams** (claimed by the manufacturer).

So our hypotheses are:

$H_0: \mu = 50$ (no difference) $H_0: \mu = 50$ $H_1: \mu \neq 50$ (mean is different) $H_1: \mu \neq 50$

We'll use the **Z-test** for this.

2. Formula for Z-statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Where:

- $\bar{X}$ = sample mean
- μ_0 = population mean (hypothesized mean)
- σ = sample standard deviation
- n = sample size

3. Python Code

```
import numpy as np
from scipy import stats

# Sample data
sample_data = [49.1, 50.2, 51.0, 48.7, 50.5, 49.8, 50.3, 50.7, 50.2, 49.6,
               50.1, 49.9, 50.8, 50.4, 48.9, 50.6, 50.0, 49.7, 50.2, 49.5,
               50.1, 50.3, 50.4, 50.5, 50.0, 50.7, 49.3, 49.8, 50.2, 50.9,
               50.3, 50.4, 50.0, 49.7, 50.5, 49.9]

# Hypothesized population mean
mu_0 = 50

# Compute statistics
mean_sample = np.mean(sample_data)
std_sample = np.std(sample_data, ddof=1)
n = len(sample_data)

# Compute Z-statistic
z_stat = (mean_sample - mu_0) / (std_sample / np.sqrt(n))
```

```
# Compute two-tailed p-value
p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))

print(f"Sample Mean: {mean_sample:.3f}")
print(f"Sample Std Dev: {std_sample:.3f}")
print(f"Z-statistic: {z_stat:.3f}")
print(f"P-value: {p_value:.4f}")

# Significance level
alpha = 0.05

# Decision
if p_value < alpha:
    print("Reject the null hypothesis (H0). There is a significant difference.")
else:
    print("Fail to reject the null hypothesis (H0). No significant difference found.")
```

4. Expected Output (Typical Result)

```
Sample Mean: 50.08
Sample Std Dev: 0.52
Z-statistic: 0.95
P-value: 0.3400
Fail to reject the null hypothesis (H0). No significant difference found.
```

5. Interpretation

- **Z = 0.95** → The sample mean is **less than 1 standard error** away from 50.
- **p-value = 0.34 > 0.05** → Not statistically significant.
- ✓ **Conclusion:** There is **no strong evidence** that the true mean weight differs from 50g.

Question 8: Write a Python script to simulate data from a normal distribution and calculate the 95% confidence interval for its mean. Plot the data using Matplotlib.

ans

1. Concept Overview

A **normal distribution** is a continuous probability distribution that is symmetric around the mean. We can simulate random data using NumPy and calculate a **confidence interval (CI)** for its mean.

2. Formula for 95% Confidence Interval

$$CI = \bar{X} \pm z \times \frac{s}{\sqrt{n}} \quad CI = \bar{X} \pm z \times ns$$

Where:

- $\bar{X}$ = sample mean
- s = sample standard deviation
- n = sample size
- $z=1.96$ for 95% confidence

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from scipy import stats
```

Generate random data from a normal distribution

```
np.random.seed(0) # for reproducibility
```

```
data = np.random.normal(loc=50, scale=5, size=1000) # mean=50, std=5, n=1000
```

Calculate sample statistics

```
mean = np.mean(data)
```

```
std = np.std(data, ddof=1)
```

```
n = len(data)
```

Calculate 95% confidence interval

```
confidence = 0.95
```

```
z_score = stats.norm.ppf(1 - (1 - confidence)/2)
```

```
margin_of_error = z_score * (std / np.sqrt(n))
```

```
lower_bound = mean - margin_of_error
```

```
upper_bound = mean + margin_of_error
```

Display results

```
print(f"Sample Mean: {mean:.2f}")
```

```
print(f"95% Confidence Interval: ({lower_bound:.2f}, {upper_bound:.2f})")
```

Plot histogram

```
plt.hist(data, bins=30, color='skyblue', edgecolor='black')
```

```
plt.title('Normal Distribution (Mean=50, SD=5)')
```

```
plt.xlabel('Values')
```

```
plt.ylabel('Frequency')
```

Add lines for mean and confidence interval

```
plt.axvline(mean, color='red', linestyle='--', label=f'Mean = {mean:.2f}')
```

```
plt.axvline(lower_bound, color='green', linestyle='--', label='95% CI Lower')
```

```
plt.axvline(upper_bound, color='green', linestyle='--', label='95% CI Upper')
```

```
plt.legend()
```

```
plt.show()
```

Question 9: Write a Python function to calculate the Z-scores from a dataset and visualize the standardized data using a histogram. Explain what the Z-scores represent in terms of standard deviations from the mean. (Include your Python code and output in the code box below.)

1. Concept Overview

A **Z-score** (also called a *standard score*) tells us **how many standard deviations a data point is away from the mean**.

$$Z = \frac{X - \bar{X}}{s} \quad Z = \frac{X - \bar{X}}{s}$$

Where:

- X = individual data value
- $\bar{X}$ = sample mean
- s = sample standard deviation

Interpretation:

- $Z = 0 \rightarrow$ exactly at the mean
- $Z = +1 \rightarrow$ 1 standard deviation **above** the mean
- $Z = -1 \rightarrow$ 1 standard deviation **below** the mean

Z-scores are useful for comparing data on **different scales** or identifying **outliers**.

2. Python Code

```
import numpy as np
import matplotlib.pyplot as plt
```



```

# Function to calculate Z-scores
def calculate_z_scores(data):
    mean = np.mean(data)
    std = np.std(data, ddof=1)
    z_scores = (data - mean) / std
    return z_scores, mean, std

# Sample dataset
data = np.array([45, 47, 50, 52, 53, 49, 51, 48, 55, 46])

# Calculate Z-scores
z_scores, mean, std = calculate_z_scores(data)

# Display results
print(f"Mean: {mean:.2f}")
print(f"Standard Deviation: {std:.2f}")
print("Z-scores:", np.round(z_scores, 2))

# Plot histogram of Z-scores
plt.hist(z_scores, bins=8, color='lightblue', edgecolor='black')
plt.title('Histogram of Standardized Data (Z-scores)')
plt.xlabel('Z-score')
plt.ylabel('Frequency')

# Mark the mean (Z = 0)
plt.axvline(0, color='red', linestyle='--', label='Mean (Z=0)')
plt.legend()
plt.show()

```